ADVANCED STATISTICS IN PRACTICE

Implications of Advanced Statistics in Business and Research

Turning data into evidence, decisions, and discovery

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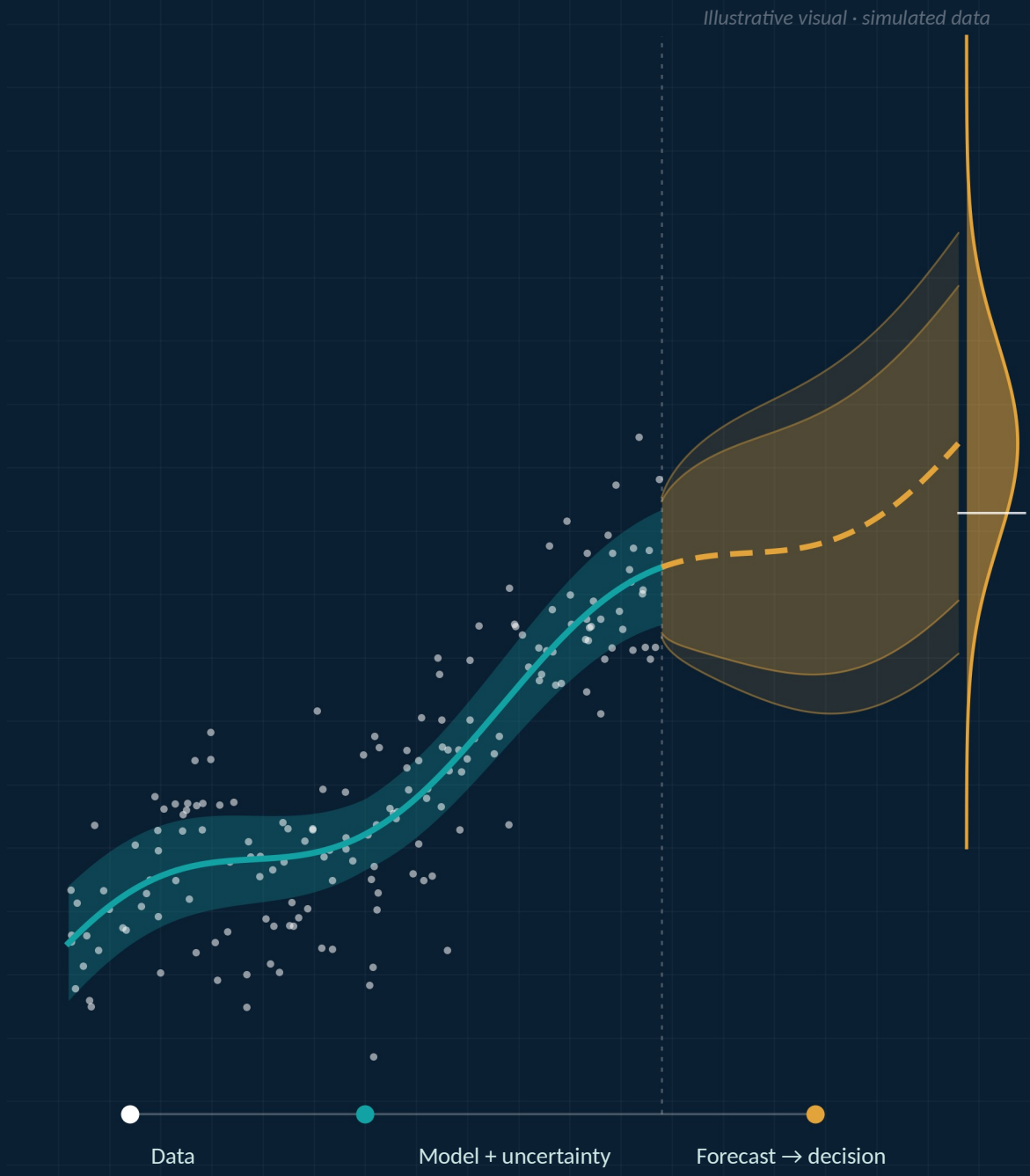
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Statistics turns data into evidence you can act on

RAW DATA

What was recorded?

Transactions, sensors, surveys and trial records — noisy and incomplete.

DESCRIPTION · BI

What happened?

Dashboards and averages summarize the past but cannot separate signal from noise.

PREDICTIVE ANALYTICS

What is likely next?

Models learn patterns to forecast outcomes and score the risk of new cases.

STATISTICAL INFERENCE

Is it real, why, and how sure are we?

Tests, intervals and causal designs weigh competing explanations.

EVIDENCE-BASED DECISION

What should we do?

Act with the expected benefit and its uncertainty made explicit.

Why intuition and simple summaries fall short

Averages hide structure

Combined data can show the opposite trend to every subgroup (Simpson's paradox).

Patterns can be noise

Without a measure of sampling variability, chance fluctuations look like real signals.

Correlation ≠ causation

A hidden confounder can drive both variables; only design or explicit causal assumptions resolve it.

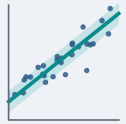
EVIDENCE FROM BUSINESS RESEARCH

5–6%

higher output and productivity in firms that use data-driven decision-making than their other investments and IT use would predict (179 large publicly traded U.S. firms).

Brynjolfsson, Hitt & Kim (2011). Observational study; instrumental variables used to address reverse causality.

Different methods answer different questions

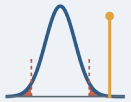


Regression & predictive modeling

Asks How does an outcome move with its drivers, and what will it be for a new case?

Applied Price elasticity and sales drivers; wage and dose–response studies.

Caveat Coefficients show association unless the design supports causation.



Hypothesis tests & confidence intervals

Asks Is a difference bigger than sampling noise, and how large could it plausibly be?

Applied Comparing campaigns or treatments; quality-control checks.

Caveat $p < 0.05$ does not prove an important effect — report effect sizes.



Bayesian inference

Asks Given prior knowledge and new data, how probable is each explanation?

Applied Updating risk estimates; adaptive clinical trials; small-sample tests.

Caveat Conclusions depend on the prior — state it and test its influence.

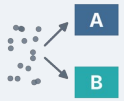


Time-series forecasting

Asks What comes next, given trend, seasonality and autocorrelation?

Applied Demand, inventory and cash-flow planning; economic and disease series.

Caveat Structural breaks can ruin forecasts; always show prediction intervals.



Experimental design & causal inference

Asks What happens if we intervene, rather than merely observe?

Applied A/B tests and pricing trials; randomized controlled trials; difference-in-differences.

Caveat Needs randomization or strong, explicitly stated assumptions.



Multivariate analysis & clustering

Asks What structure exists across many variables at once?

Applied Customer segments; survey scales (PCA, factor analysis); gene groups.

Caveat Clusters are descriptive and shift with variable scaling and the chosen k.

Choosing a method: research objective → data structure (scale, dependence, sample size) → assumptions → decision context

Statistics changes how firms decide

VERIFIED CASE Bing search ads: one online controlled experiment, from question to outcome (Kohavi & Thomke, 2017)

1 · QUESTION
Would longer ad titles raise revenue without hurting the user experience?

2 · DATA
Users randomly assigned to the old or new ad layout; revenue and user-experience metrics logged.

3 · METHOD
Online controlled experiment (A/B test) with significance tests on each metric.

4 · INSIGHT
Revenue up about 12% with no harm to key user-experience metrics; result replicated.

5 · DECISION
Ship an idea that had sat in the backlog for more than six months.

6 · OUTCOME
Estimated at more than $100 million a year in U.S. revenue at the time.

Across business functions: question, method, decision

Function	Key question	Statistical method	Decision supported
Demand & inventory	How much will we sell next quarter, and how uncertain is that?	Time-series models (ETS, ARIMA) with prediction intervals	Set orders and safety stock from the interval, not a single point guess
Customer segmentation	Which customers behave alike, and how do the groups differ?	Cluster analysis; principal component analysis	Tailor offers, channels and product design to each segment
Pricing & revenue	How does demand respond to a price change?	Price-elasticity regression; randomized price tests	Change prices with an expected revenue effect and its uncertainty
Credit risk & fraud	How likely is default or fraud for this application?	Logistic-regression scorecards; anomaly detection	Approve, price or flag — with error rates and fairness monitored
Churn & retention	Who will leave, and does an offer change that?	Survival or classification models plus uplift experiments	Target customers the offer persuades, not just those most at risk

Rigor lives in the design, not just the analysis

1 **Research question**
Testable hypothesis; primary outcome fixed in advance

2 **Study design**
Randomized or observational? Plan for confounding

3 **Data collection**
Sample size from a power analysis; valid measures

4 **Statistical analysis**
Model matched to the data and its assumptions

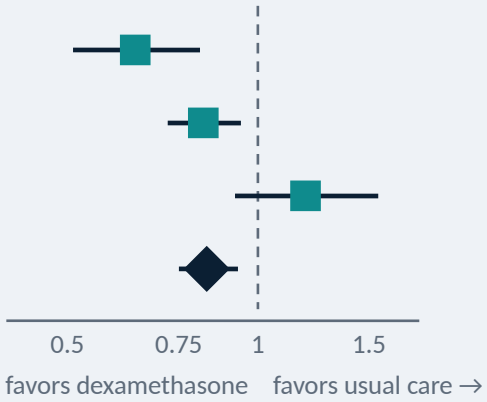
5 **Interpretation**
Effect sizes, confidence intervals, robustness checks

6 **Conclusion**
Claims proportional to evidence; reproducible code and data

EVIDENCE EXAMPLE **RECOVERY randomized trial: dexamethasone vs. usual care**

28-day mortality, hospitalized COVID-19 patients · 6,425 randomized (2,104 vs. 4,321) · age-adjusted rate ratios

Respiratory support	Dexa.	Usual care	Rate ratio (95% CI)
Invasive ventilation	29.3%	41.4%	0.64 (0.51–0.81)
Oxygen only	23.3%	26.2%	0.82 (0.72–0.94)
No respiratory support	17.8%	14.0%	1.19 (0.92–1.55)
All patients	**22.9%**	**25.7%**	**0.83 (0.75–0.93)**



Log scale. Redrawn from RECOVERY, NEJM 2021.

REPRODUCIBILITY CHECK

97% → 36%

of 100 published psychology studies reported significant results; only 36% of their high-powered replications did. Replication effects were about half as large.

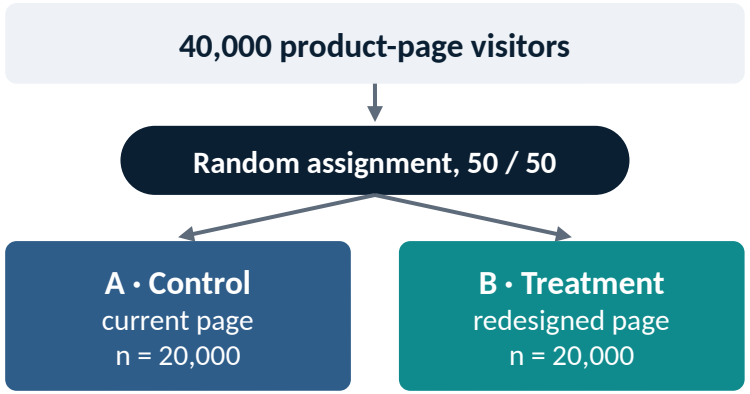
Open Science Collaboration (2015), Science

Low power, flexible analysis and selective publication inflate findings; pre-registration and open data counter them.

ILLUSTRATIVE DATA hypothetical, not real company results

Does a redesigned product page lift conversion?

1 Design



Hypotheses $H_0: p_B = p_A$ $H_1: p_B \neq p_A$

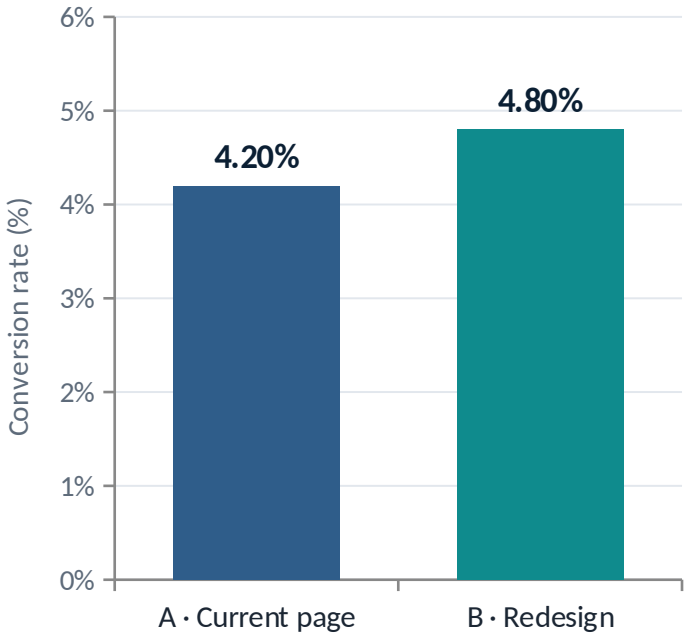
Test two-sided, $\alpha = 0.05$

Primary metric conversion rate = orders ÷ visitors

Power 80% to detect +0.6 pp from a 4.2% baseline needs ≈ 18,740 visitors per arm → 20,000 planned

Discipline fixed test length set in advance; no stopping on interim peeks

2 Results



A: 840 orders / 20,000 visitors = 4.20%
B: 960 orders / 20,000 visitors = 4.80%

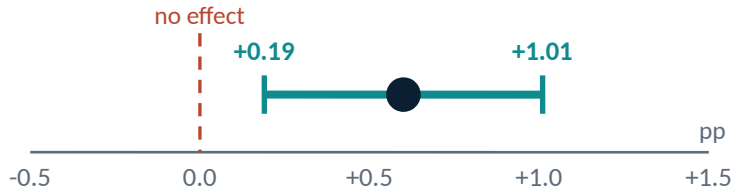
Axis starts at zero. Separate 95% CIs:
A 3.92–4.48% · B 4.50–5.10%

3 Inference & decision

Method two-proportion z-test (large samples, independent visitors, random assignment)

+0.60 pp	+14.3%	p = 0.004
difference (B − A)	relative lift	z = 2.89, two-sided

95% confidence interval for the difference



If the redesign had no effect, a gap this large would arise in only about 0.4% of repeated experiments. The true lift is plausibly anywhere from +0.19 to +1.01 pp: likely real, but its size is uncertain.

DECISION Roll out B if guardrail metrics (returns, page speed) show no harm; keep monitoring for novelty effects.

Sophistication without discipline misleads

Risk	What goes wrong	Safeguard
Data quality & sampling bias	Sample does not represent the population of interest	Audit data provenance; reweight or redesign the sample
Violated model assumptions	Biased estimates and wrong standard errors	Residual diagnostics; robust or alternative models
Overfitting	Model memorizes noise and fails on new data	Hold-out data and cross-validation; simpler models
Multiple testing & p-hacking	Testing many outcomes guarantees some false positives	Pre-registration; Holm or false-discovery-rate correction
Confounding	Spurious causal claims from observational data	Randomize, or use explicit causal designs and diagrams
Misreading significance	"Significant" read as "large" or "important"	Report effect sizes with confidence intervals
Privacy & fairness	Models harm individuals or disadvantage groups	Consent, data minimization, bias audits across groups
Missing domain expertise	Valid statistics answer the wrong question	Involve domain experts; report methods transparently

MACHINE LEARNING & AI

ML excels at prediction from large, complex data and increasingly complements statistical modeling. But accurate prediction does not show what would happen under an intervention: prediction is not causation (Shmueli, 2010).

Example: a widely used U.S. health algorithm predicted cost as a proxy for need, and so understated the needs of Black patients (Obermeyer et al., 2019).

RESPONSIBLE PRACTICE

1. **Plan** pre-specify question, outcome, analysis
2. **Check** data quality and model assumptions
3. **Estimate** effect sizes with their uncertainty
4. **Stress-test** robustness and hold-out data
5. **Report** transparently, limitations included

Reliable conclusions are where data creates value

01 **Advanced statistics turns raw data into structured evidence.**
Description, prediction and inference answer different questions.

02 **Businesses decide better by quantifying uncertainty and testing alternatives.**
Experiments and intervals replace opinion with measured risk.

03 **Researchers strengthen conclusions through design and rigorous inference.**
Power, randomization and robustness checks matter as much as models.

04 **Responsible interpretation matters as much as technical sophistication.**
Effect sizes, transparency and fairness keep results trustworthy.

> *The value of data lies not merely in collecting it, but in drawing reliable conclusions from it.*

Thank you · Questions and discussion

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Full reference list with DOIs and visual sources: Appendix A (supplementary slide).

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Visual sources All charts and diagrams are original to this presentation. Title visual and method diagrams use simulated data (illustrative). Slide 5 forest plot is redrawn from published RECOVERY values. Slide 6 data are hypothetical. No third-party images are used.